```
Custom symbols registered.
```

```
  Cell In[2], line 12
    conc create_concrete(fck=f_ck.value)
         ^
SyntaxError: invalid syntax
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[3], line 1
----> 1 f_cd = conc.fcd() * MPa  # Convert back to unit-aware variable
      3 A_c = b * h
      4 A_s = n_bars * (pi * (bar_dia/2)**2)

NameError: name 'conc' is not defined
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 sec['properties']
      2 print(sec['edi_std'])

NameError: name 'sec' is not defined
```

```
Total entries across all libraries: 49625
```

$$A = 62000.000 \ \ (\mathrm{mm2})$$

$$\mathrm{Iy} = 10218840000.000 \ \ (\mathrm{mm4})$$

$$\mathrm{Iz} = 267210000.000 \ \ (\mathrm{mm4})$$

$$d = 1036.300 \ \ (\mathrm{mm})$$

$$\mathrm{bf} = 308.500 \ \ (\mathrm{mm})$$

$$\mathrm{tf} = 54.100 \ \ (\mathrm{mm})$$

$$\mathrm{tw} = 30.000 \ \ (\mathrm{mm})$$

```
ry (mm) = 405.98029508832076
rz (mm) = 65.64936183754888
```

	library	label	type	designation	D	BF	TF	TW	A
0	BSShapes2006	UKB1016X305X487	STEEL_I_SECTION	W	1036.3	308.5	54.1	30.0	62000.0
1	BSShapes2006	UKB1016X305X437	STEEL_I_SECTION	W	1026.1	305.4	49.0	26.9	55700.0
2	BSShapes2006	UKB1016X305X393	STEEL_I_SECTION	W	1015.9	303.0	43.9	24.4	50000.0
3	BSShapes2006	UKB1016X305X349	STEEL_I_SECTION	W	1008.1	302.0	40.0	21.1	44500.0
4	BSShapes2006	UKB1016X305X314	STEEL_I_SECTION	W	999.9	300.0	35.9	19.1	40000.0
5	BSShapes2006	UKB1016XX305X272	STEEL_I_SECTION	W	990.1	300.0	31.0	16.5	34700.0
6	BSShapes2006	UKB1016X305X249	STEEL_I_SECTION	W	980.1	300.0	26.0	16.5	31700.0
7	BSShapes2006	UKB1016X305X222	STEEL_I_SECTION	W	970.3	300.0	21.1	16.0	28300.0
8	BSShapes2006	UKB914X419X388	STEEL_I_SECTION	W	921.0	420.5	36.6	21.4	49400.0
9	BSShapes2006	UKB914X419X343	STEEL_I_SECTION	W	911.8	418.5	32.0	19.4	43700.0
10	BSShapes2006	UKB914X305X289	STEEL_I_SECTION	W	926.6	307.7	32.0	19.5	36800.0
11	BSShapes2006	UKB914X305X253	STEEL_I_SECTION	W	918.4	305.5	27.9	17.3	32300.0
12	BSShapes2006	UKB914X305X224	STEEL_I_SECTION	W	910.4	304.1	23.9	15.9	28600.0
13	BSShapes2006	UKB914X305X201	STEEL_I_SECTION	W	903.0	303.3	20.2	15.1	25600.0
14	BSShapes2006	UKB838X292X226	STEEL_I_SECTION	W	850.9	293.8	26.8	16.1	28900.0
15	BSShapes2006	UKB838X292X194	STEEL_I_SECTION	W	840.7	292.4	21.7	14.7	24700.0
16	BSShapes2006	UKB838X292X176	STEEL_I_SECTION	W	834.9	291.7	18.8	14.0	22400.0
17	BSShapes2006	UKB762X267X197	STEEL_I_SECTION	W	769.8	268.0	25.4	15.6	25100.0
18	BSShapes2006	UKB762X267X173	STEEL_I_SECTION	W	762.2	266.7	21.6	14.3	22000.0
19	BSShapes2006	UKB762X267X147	STEEL_I_SECTION	W	754.0	265.2	17.5	12.8	18700.0

General calculation

Calculation Report

Author: Vance Kang **Date:** {{}}
Project No.:

Calculations

Simply supported beam with UDL

$L = 13$ (Span length, m) $w = 0.209$ (UDL, kN/m) $E = 200000000.000$ (Y

$I = 0.000$ (Second moment of area, m^4) $x = 6.500$ (Position along beam, m)

My Image

$$R = w \cdot \frac{L}{2} = 0.209 \cdot \frac{13}{2} = 1.359$$

$$M_{max} = w \cdot \frac{(L)^2}{8} = 0.209 \cdot \frac{(13)^2}{8} = 4.415$$

$$\delta_{max} = 5 \cdot w \cdot \frac{(L)^4}{384 \cdot E \cdot I} = 5 \cdot 0.209 \cdot \frac{(13)^4}{384 \cdot 200000000.000 \cdot 0.000} = 0.055$$

---

simply supported with 1 concentrated loads

$L = 6.600$ (span length, ℓ (m)) $P = 40$ (each point load (kN)) $P_{ult} = 72.000$ (factored load (kN))

$$R = \frac{P_{ult}}{2} = \frac{72.000}{2} = 36.000$$

$$M_{max} = P_{ult} \cdot \frac{L}{4} = 72.000 \cdot \frac{6.600}{4} = 118.800$$

---

simply supported with 2 concentrated loads

$L = 8.300$ (span length, ℓ (m))

$P = 0.900$ (each point loac

$E = 200000000.000$ (Young's modulus (kN/m^2) 200 GPa = 200e6 kN/m^2)

$I = 0.000$ (second momen

$x_{mid} = 4.150$ (a point with a < x < (L - a) (m))

My Image

$R = 0.900$ (R = V = P)

$M_{max} = P \cdot a = 0.900 \cdot 3.500$ $= 3.150$ (Maximum moment between lo

$$\delta_{max} = \frac{P \cdot a}{24 \cdot E \cdot I} \cdot \left(3 \cdot (L)^2 - 4 \cdot (a)^2\right)$$
$$= \frac{0.900 \cdot 3.500}{24 \cdot 200000000.000 \cdot 0.000} \cdot \left(3 \cdot (8.300)^2 - 4 \cdot (3.500)^2\right)$$
$$= 0.001 \text{ (Maximum deflection at center)}$$

Grating Loading Calculation

$\rho = 80$ (steel grating area load assumed (kg/m2))

$w_{DL} = \rho \cdot 10 \cdot (10)^{-3} = 80 \cdot 10 \cdot (10)^{-3}$

$\text{LL} = 2.500$ (assumed live load)

$\text{LW} = 1$ (load width(m))

$q = (1.35 \cdot w_{DL} + 1.5 \cdot \text{LL}) \cdot \text{LW} = (1.35 \cdot 0.800 + 1.5 \cdot 2.500) \cdot 1$ $= 4.8$

load *per anchor* $= \frac{q}{\frac{1000}{300}} = \frac{4.830}{\frac{1000}{300}}$ $= 1.449$ ((KN) spaced eve

Portal frame Moment

My Image

$w = 10.000$ (projected loading kN/m) $L = 10.000$ (Total span m) $I_R = 1.($

$I_C = 1.500$ (preliminary assumption from the note) $h = 3.000$ (height up to lowest rafter m) $f = 2$

$s = 5.385$ (length of rafter m)

$$s = \sqrt{\left(\frac{L}{2}\right)^2 + (f)^2} = \sqrt{\left(\frac{10.000}{2}\right)^2 + (2)^2} \qquad = 5.385 \text{ (length of rafter m)}$$

$$k = \frac{I_R \cdot h}{I_C \cdot s} = \frac{1.000 \cdot 3.000}{1.500 \cdot 5.385} \qquad = 0.371$$

$$\phi = \frac{f}{h} = \frac{2}{3.000} \qquad = 0.667$$

$$m = 1 + \phi = 1 + 0.667 \qquad = 1.667$$

$$B = 2 \cdot (k + 1) + m = 2 \cdot (0.371 + 1) + 1.667 \qquad = 4.409$$

$$C = 1 + 2 \cdot m = 1 + 2 \cdot 1.667 \qquad = 4.333$$

$$N = B + m \cdot C = 4.409 + 1.667 \cdot 4.333 \qquad = 11.632$$

$$M_E = w \cdot (L)^2 \cdot \frac{3 + 5 \cdot m}{16 \cdot N} = 10.000 \cdot (10.000)^2 \cdot \frac{3 + 5 \cdot 1.667}{16 \cdot 11.632} \qquad = 60.897$$

$$M_A = w \cdot \frac{(L)^2}{8} + m \cdot M_E = 10.000 \cdot \frac{(10.000)^2}{8} + 1.667 \cdot 60.897 \qquad = 226.495$$

$$M_A = w \cdot \frac{(L)^2}{8} = 10.000 \cdot \frac{(10.000)^2}{8} \qquad = 125.000$$